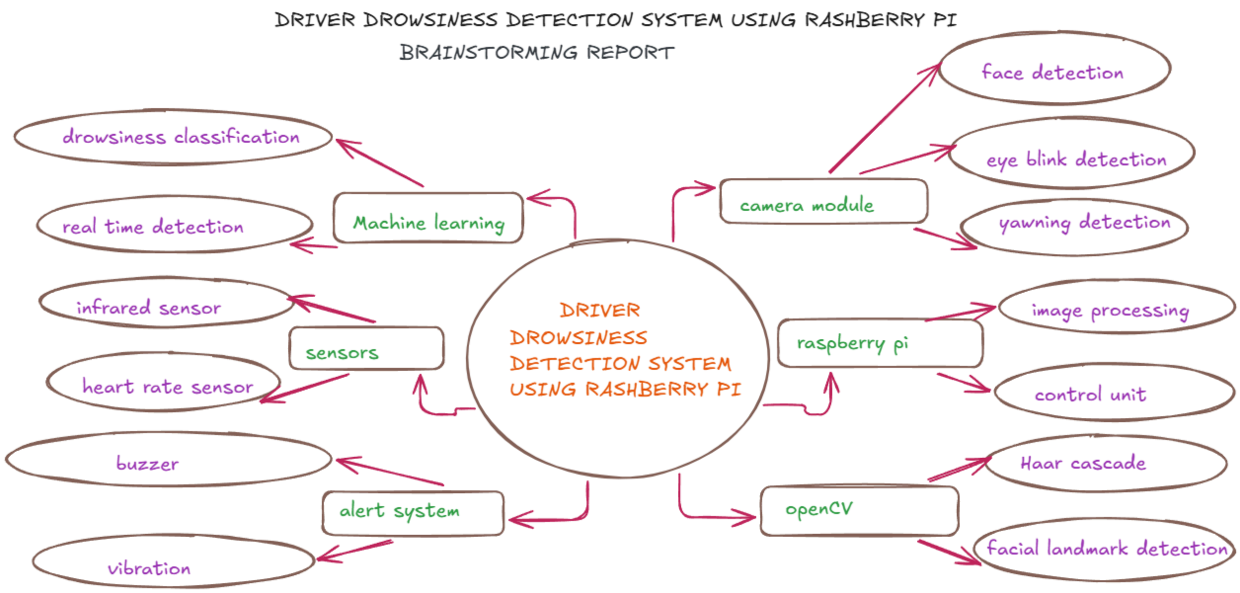
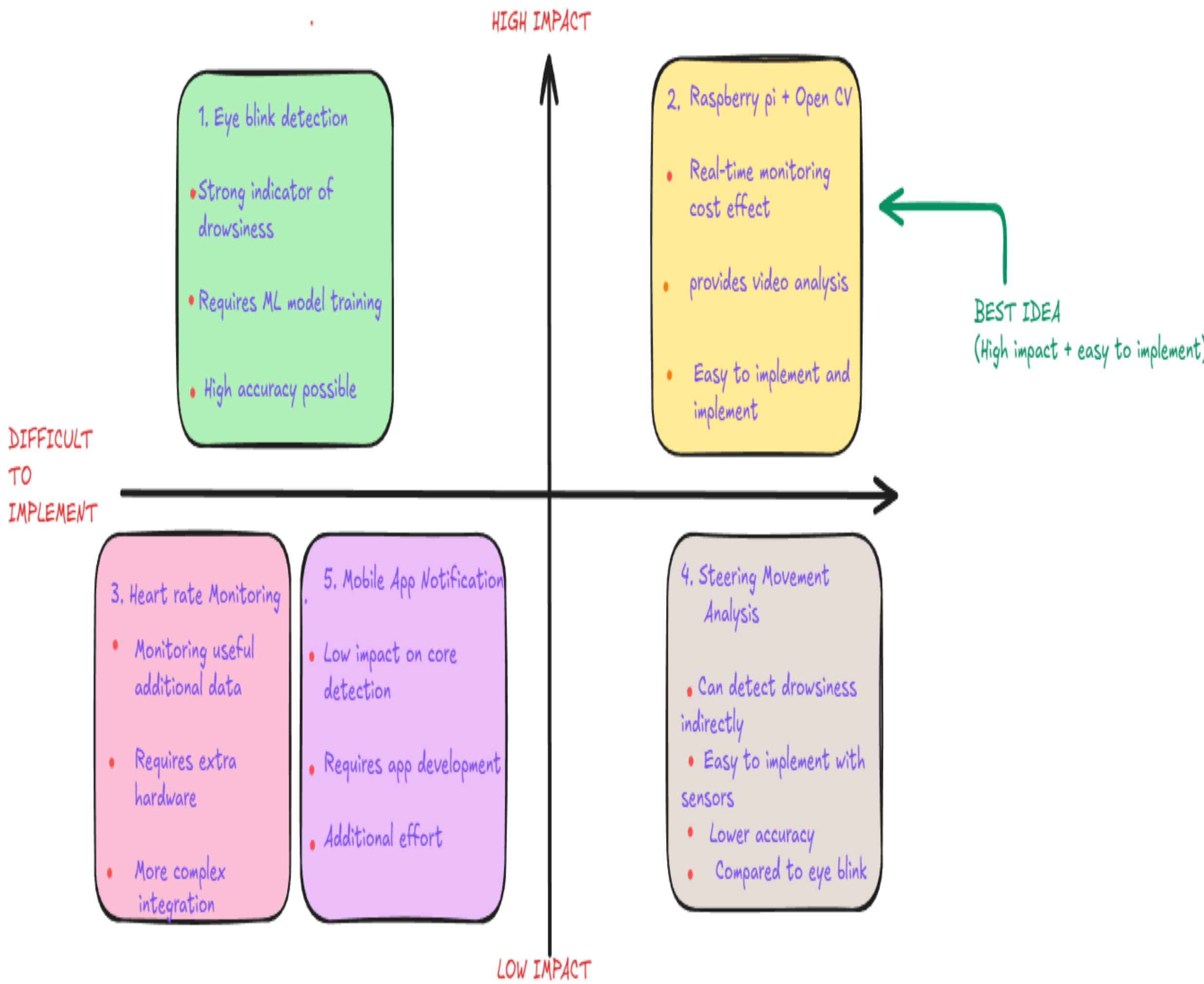
TEM MEMBER
LAVANYA M
ANUJASRI T
DHARSHINI S
PREETHIKA S



DRIVER DROWSINESS DETECTION SYSTEM USING RASPBERRY PI

IDEA PRIORITIZATION MATRIX



DRIVER DROWSINESS DETECTION SYSTEM USING RASPBERRY PI
EMPATHY MAP

SAYS
- I feel sleepy while driving long hours.
- I need an alert when I start losing focus.
- I want a system that is not distracting.

THINKS
- Safety is important for me and others.
- I want to reach my destination safely.
- I hope the system is accurate and reliable.

DRIVER

FEELS
- Tired
- Stressed
- Worried about accidents
- Responsible for passengers

Does
- Drivers for long hours
- Tries to stay awake(e.g., drinks coffee, opens window).
- Ignores early signs of drowsiness

GOALS
- Real-time drowsiness detection
- Non-intrusive alerts
- Affordable and reliable system

PAIN POINTS
- No early warning for drowsiness
- Hard to self-detect drowsiness
- Existing systems are expensive
- Distraction from phone alerts

DRIVER DROWSINESS DETECTION SYSTEM USING RASHBERRY PI
USER PERSONA

Age	28
Occupation	software Engineer
Location	Chennai, India
Driving type	Daily commuter
Driving hours	1-2 hours(per day) long drives on weekends
Health	normal
Device usage	Smartphone, laptop, Smartwatch

"I want a system that helps me stay alert and safe while driving, without being distracting."

ABOUT

Arun is a software engineer who travels daily to office and occasionally goes on long road trips. He values safety and wants to avoid fatigue while driving. He is tech-savvy and open to using smart systems that can alert him in real time.

GOALS

Stay alert and safe while driving.
Avoid accidents caused by drowsiness
Get real-time alerts when feeling sleepy
Trustworthy, accurate and affordable system
Easy to install and use in any car

FRUSTRATIONS

Feels sleepy during long driver especially at night
Yawning and eye closing goes unnoticed
Existing solutions are expensive or not user friendly
Frequent phone notification cause distraction

SCENARIO

Arun starts driving home after a long day at work. during the drive, he begins to feel sleepy. the system detects his drowsiness(eye closure & yawning) and alerts him immediately with a buzzer. He takes a break and continues driving safely.

NEEDS & EXPECTATION

Real-time monitoring of eyes and facial behavior
Immediate alert(sound/buzzer /LED/vibration)
Affordable and reliable
Works well in low lightcondition
Non-intrusive and distraction free system
compact system installedeasily

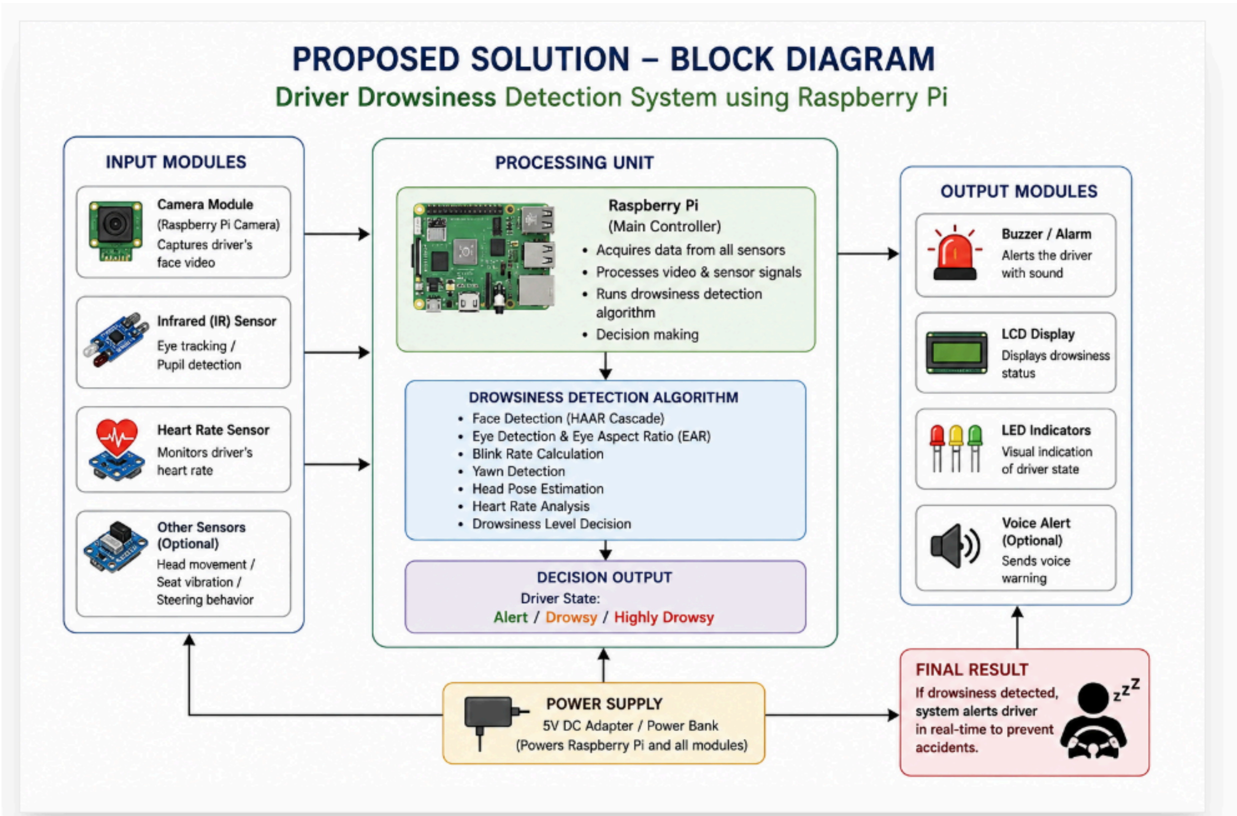
BEHAVIOR & HABITS

Drives to office five days a week
Often takes late-night drives on weekends
Takes short breaks during long drives
uses tech gadgets and smart solutions
Health conscious, safety

PAIN POINTS

Drowsiness leads to slow reaction and accidents
hard to self-detect drowsiness
No affordable, real time and accurate solution available
Distraction from mobile notifications increases risk

Proposed solution



Functional requirement

S.No	Functional requirement	Description
1	Driver face detection	The system shall capture and detect the driver's face using a camera.
2	Eye blink detection	The system shall detect eye blinking and eye closure patterns.
3	Yawning detection	The system shall detect yawning to identify drowsiness.
4	Head movement detection	The system shall monitor abnormal head movements indicating drowsiness.
5	Heart rate monitoring	The system shall monitor the driver's heart rate using a sensor.
6	Drowsiness Analysis	The system shall analyze facial features and sensor data using openCV and machine learning.
7	Alert generation	The system shall activate a buzzer or vibration alert when drowsiness is detected.
8	Continuous Monitoring	The system shall continuously monitor the driver while the vehicle is operating.

Non functional requirement

S.No	Non functional requirement	description
1	Accuracy	The system should detect drowsiness with high accuracy.

2	Real-time performance	The system should provide alerts without noticeable delay.
3	Reliability	The system should operate continuously without frequent failures.
4	Low Cost	The system should use affordable components such as raspberry pi.
5	Low power consumption	The system should consume minimal electrical power.
6	Ease of use	The system should be simple to install and operate.
7	Scalability	The system should be adaptable for different types of vehicles.
8	Safety	The system should not distract the driver during normal operation.

Project planning

Gantt Chart

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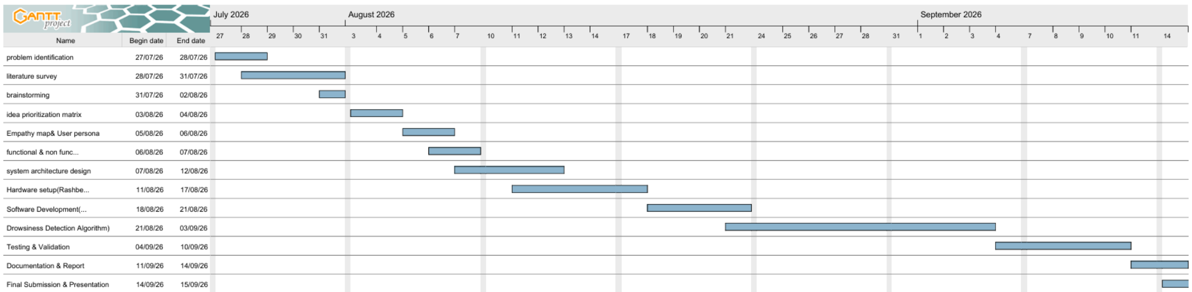

Risk Analysis

Objective: To identify potential risks that may affect the successful development and performance of the system and to plan mitigation strategies.

Risk ID	Risk Description	Category	Probability(Low/Med/High/)	Impact(Low/Med/High)	Risk Level(PxI)	Mitigation Strategy
R1	**Poor Lighting Conditions** Low or excessive lighting may affect face eye detection accuracy.	Technical	Medium	High	High	• Use an IR(infrared) camera or IR LEDs for night vision. • Apply image enhancement techniques.
R2	**Sensor Inaccuracy** Heart rate					• Use reliable and calibrated sensors.

	or eye blink sensors may give incorrect or noisy data.	Technical	Low	Medium	Medium	• Implement data filtering and validation.
R3	**Algorithm Errors** Errors in code or wrong algorithm selection may lead to false alerts or missed detections.	Technical	Medium	High	High	• Test multiple algorithms. • Validate using a large dataset and real-time testing.
R4	**System Performance** High CPU usage or low processing power of Raspberry Pi may cause lag.	Technical	Medium	Medium	Medium	• Optimize code. • Use lightweight models and efficient libraries.
R5	**Power Failure** Unexpected power loss may stop the system during	Hardware	Low	High	Medium	• Use a stable power supply. • Consider a backup battery(UPS/Power bank).

	operation.					
R6	**User Non-compliance** Drivers may ignore alerts or disable the system.	Human	Medium	Medium	Medium	● Provide clear audio-visual alerts. ● Educate users about the importance of the system.
R7	**Hardware Failure** Camera or sensors may fail or get damaged.	Hardware	Low	Medium	Low	● Use quality components. ● Regular maintenance and testing.